

13. In a mixture of two monatomic ideal gases X and Y in thermal equilibrium, a molecule of Y has twice the mass of a molecule of X. The mean random translational kinetic energy of the molecules of Y is 6.0×10^{-21} J.

What is the mean random translational kinetic energy of the molecules of X?

- A 3.0×10^{-21} J
- B 4.2×10^{-21} J
- C** 6.0×10^{-21} J
- D 12×10^{-21} J

Answer: C

- X and Y are in thermal equilibrium, so thermodynamic temperature is the same
- since mean k.e. per molecule is proportional to temperature, mean k.e. of molecules of X is the same as Y